

Supplemental Material for “Exact classification of moments through second order for deterministic reset maps of an inertial active particle”

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This supplement gives the generator, closed second-order hierarchy, exhaustive pairwise proof, restricted-identity ledger, and mandatory limits for the two-dimensional inertial active Brownian particle with deterministic Poisson resetting. It is written to be read independently. Repository links are included only as traceability pointers to the accepted symbolic records and implementations.

1 S1. Model, reset semantics, and generator

Let $r = (x, y)^\top$, $v = (v_x, v_y)^\top$, and $u(\theta) = (\cos \theta, \sin \theta)^\top$. Between resets,

$$dr = v dt, \quad M dv = -(v - \text{Pe } u) dt + \sqrt{2} dW_t, \quad d\theta = \sqrt{2} dW_{r,t},$$

where W_t is a standard two-component Wiener process, $W_{r,t}$ is an independent scalar Wiener process, and

$$M > 0, \quad \text{Pe} \geq 0, \quad \rho \geq 0.$$

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Reset times are those of an independent Poisson process of rate ρ . Each reset is instantaneous. With 0 denoting the appropriate scalar or two-component zero, the seven nonempty deterministic maps are

Protocol q	Full post-reset state $R_q(r, v, \theta)$
P	$(0, v, \theta)$
V	$(r, 0, \theta)$
Θ	$(r, v, 0)$
PV	$(0, 0, \theta)$
$P\Theta$	$(0, v, 0)$
$V\Theta$	$(r, 0, 0)$
$PV\Theta$	$(0, 0, 0)$

Every unlisted state variable is retained exactly. For a smooth observable $f(r, v, \theta)$, the backward generator is

$$\begin{aligned} \mathcal{L}_q f = & v \cdot \nabla_r f - \frac{v - \text{Pe} u}{M} \cdot \nabla_v f + \frac{1}{M^2} \Delta_v f + \partial_\theta^2 f \\ & + \rho [f(R_q(r, v, \theta)) - f(r, v, \theta)]. \end{aligned}$$

The common default initial state is $r(0) = 0$, $v(0) = 0$, and $\theta(0) = 0$. Thus $\bar{r}(0) = \bar{v}(0) = 0$, $\bar{u}(0) = e = (1, 0)^\top$, all raw second moments containing r or v initially vanish, and $U(0) = ee^\top$.

Traceability: [scientific contract](#) and [symbolic generator implementation](#).

2 S2. Temporal scope of the second-order record

For a protocol q , define

$$\mathcal{R}_2(q; M, \text{Pe}, \rho)$$

to be the fixed stationary-or-asymptotic component-level output used by the classification. For a position-reset map, the record contains the stationary long-time limits of all means and raw second moments below, the centered spatial covariance, and the finite stationary variance. For a map without position reset, the record contains the stationary internal (v, u) moments and the accepted long-time spatial polynomial coefficients: $\bar{r}(t) = \bar{v}t + b + o(1)$, $\text{Cov} r(t) = 2Dt + O(1)$, the corresponding raw R, C, Q coefficients, D_{eff} , and the centered and raw spatial-growth classes. Equality means equality of every applicable stationary value and every applicable long-time coefficient at matched (M, Pe, ρ) . The theorem does not assert equality or distinguishability of unspecified finite-time transients, and it does not claim finite-sample recoverability.

The raw matrices used throughout are

$$\begin{aligned} R &= E[rr^\top], & S &= E[vv^\top], & C &= E[rv^\top], \\ Q &= E[ru^\top], & W &= E[vu^\top], & U &= E[uu^\top], \end{aligned}$$

with means $\bar{r} = E[r]$, $\bar{v} = E[v]$, and $\bar{u} = E[u]$. The required named scalar contractions are

$$\text{TrR} := \text{Tr } R = E|r|^2, \quad \text{TrS} := \text{Tr } S = E|v|^2, \quad \text{TrC} := \text{Tr } C = E[r \cdot v].$$

The centered spatial covariance is $\text{Cov}(r) = R - \bar{r}\bar{r}^\top$. For a diffusive protocol,

$$D_{\text{eff}} = \lim_{t \rightarrow \infty} \frac{\text{Tr Cov}(r(t))}{4t}.$$

3 S3. Ordered redundant 28-coordinate frame and exact linear moment hierarchy

The implementation stores the ordered coordinate frame

$$\begin{aligned} \mathcal{F}_2 = & (1; \\ & r_x, r_y; \\ & v_x, v_y; \\ & u_x, u_y; \\ & R_{xx}, R_{xy}, R_{yy}; \\ & S_{xx}, S_{xy}, S_{yy}; \\ & C_{xx}, C_{xy}, C_{yx}, C_{yy}; \\ & Q_{xx}, Q_{xy}, Q_{yx}, Q_{yy}; \\ & W_{xx}, W_{xy}, W_{yx}, W_{yy}; \\ & U_{xx}, U_{xy}, U_{yy}). \end{aligned}$$

These are 28 stored coordinates, not an unrestricted 28-element basis. Every physical moment record lies on the invariant manifold

$$\boxed{\text{one} = 1, \quad \text{Tr } U = U_{xx} + U_{yy} = 1,}$$

or, in homogeneous coordinate form, $\text{one} - U_{xx} - U_{yy} = 0$. The symmetric matrices R, S, U retain three stored components, while the nonsymmetric matrices C, Q, W retain all four. Together with the constant and six first moments, the implementation inventory is $1 + 6 + 9 + 12 = 28$ entries. The exact trace constraint makes this inventory a redundant moment frame. Retaining it keeps the accepted generator, formula, and serialization interfaces stable.

In implementation-stable names, the same exact order is

```
one,
r_x, r_y,
v_x, v_y,
u_x, u_y,
rr_xx, rr_xy, rr_yy,
vv_xx, vv_xy, vv_yy,
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rv_xx, rv_xy, rv_yx, rv_yy,
ru_xx, ru_xy, ru_yx, ru_yy,
vu_xx, vu_xy, vu_yx, vu_yy,
uu_xx, uu_xy, uu_yy

Here $\mathbf{rr}$, $\mathbf{vv}$, $\mathbf{rv}$, $\mathbf{ru}$, $\mathbf{vu}$, and $\mathbf{uu}$ denote R, S, C, Q, W, U , respectively.

Write the reset indicators of protocol q as $(p, \nu, \eta) \in \{0, 1\}^3$, for position, velocity, and orientation reset, respectively. For binary indicators, write $a \vee b = a + b - ab$. Direct application of $\mathcal{L}_q$ gives the exact closed hierarchy

$$\dot{\mathbf{i}} = 0,$$

$$\begin{aligned}\dot{\bar{u}} &= -\bar{u} + \rho\eta(e - \bar{u}), \\ \dot{U} &= 2\text{Tr}(U)I - 4U + \rho\eta(ee^\top - U),\end{aligned}$$

This trace form is the exact coordinate representation used by the 28-component closure: $\dot{U}_{xx} = -2U_{xx} + 2U_{yy} + \rho\eta(1 - U_{xx})$, $\dot{U}_{xy} = -(4 + \rho\eta)U_{xy}$, and $\dot{U}_{yy} = 2U_{xx} - 2U_{yy} - \rho\eta U_{yy}$. The physical invariant $\text{Tr} U = 1$ then also permits the equivalent shorthand $2I - 4U$ on admissible states. Indeed, taking the trace gives

$$\frac{d}{dt}(\text{Tr} U - 1) = -\rho\eta(\text{Tr} U - 1).$$

Protocols without orientation reset ($\eta = 0$) preserve the trace constraint, while protocols with orientation reset ($\eta = 1$) restore any trace deviation exponentially at rate ρ . Thus every one of the seven generators leaves the physical invariant manifold unchanged. The generator's explicit physical-state validation entry point rejects states outside this manifold rather than assigning them a physical interpretation.

$$\begin{aligned}\dot{\bar{v}} &= -\frac{\bar{v} - \text{Pe} \bar{u}}{M} - \rho\nu\bar{v}, \\ \dot{W} &= \frac{\text{Pe}}{M}U - \left(1 + \frac{1}{M}\right)W - \rho(\nu \vee \eta)W + \rho\eta(1 - \nu)\bar{v}e^\top, \\ \dot{S} &= -\frac{2}{M}S + \frac{\text{Pe}}{M}(W + W^\top) + \frac{2}{M^2}I - \rho\nu S,\end{aligned}$$

and

$$\begin{aligned}\dot{\bar{r}} &= \bar{v} - \rho p \bar{r}, \\ \dot{Q} &= W - Q - \rho(p \vee \eta)Q + \rho\eta(1 - p)\bar{r}e^\top, \\ \dot{C} &= S - \frac{1}{M}C + \frac{\text{Pe}}{M}Q - \rho(p \vee \nu)C, \\ \dot{R} &= C + C^\top - \rho p R.\end{aligned}$$

These tensor equations are precisely the component equations $\dot{m}_q = A_q m_q$ in the ordered frame $\mathcal{F}_2$. The constant entry carries the affine terms. No moment outside $\mathcal{F}_2$ is generated. Any uniqueness statement for this coordinate system is restricted to the physical invariant manifold above; no unrestricted ambient-space uniqueness is claimed.

3.1 S3.1 Long-time solution for position-reset protocols

For $P, PV, P\Theta, PV\Theta$ with $\rho > 0$, every position block has a strict decay contribution. The hierarchy is triangular from $(\bar{u}, U)$ through $(\bar{v}, W, S)$ to $(\bar{r}, Q, C, R)$, so it has a unique finite stationary solution on the physical invariant manifold.

The orientation block is

q	$\bar{u}$	U
P, PV	0	$I/2$
$P\Theta, PV\Theta$	$\frac{\rho}{1+\rho}e$	$\text{diag}\left(\frac{\rho+2}{\rho+4}, \frac{2}{\rho+4}\right)$.

The velocity blocks are

q	$\bar{v}$	W
P	$\text{Pe } \bar{u}$	$\frac{\text{Pe } U}{1+M}$
PV	$\frac{\text{Pe } \bar{u}}{1+M\rho}$	$\frac{\text{Pe } U}{1+M(1+\rho)}$
$P\Theta$	$\text{Pe } \bar{u}$	$\frac{\text{Pe } U + M\rho \bar{v}e^\top}{1+M(1+\rho)}$
$PV\Theta$	$\frac{\text{Pe } \bar{u}}{1+M\rho}$	$\frac{\text{Pe } U}{1+M(1+\rho)}$.

For all four protocols,

$$\begin{aligned}\bar{r} &= \frac{\bar{v}}{\rho}, \\ S &= \frac{\text{Pe}(W + W^\top) + 2I/M}{2 + M\rho \mathbf{1}_{\{V \text{ reset}\}}}, \\ Q &= \frac{W}{1+\rho}, \\ C &= \frac{MS + \text{Pe } Q}{1+M\rho}, \\ R &= \frac{C + C^\top}{\rho}.\end{aligned}$$

Consequently, all four position-reset protocols are stationary/localized in centered position and have bounded raw MSD.

Traceability: [accepted position-protocol solution](#).

3.2 S3.2 Long-time solution for non-position protocols

The internal (v, u) blocks of $V, \Theta, V\Theta$ are exactly the same as those of $PV, P\Theta, PV\Theta$, respectively. Define

$$\Sigma_u = U - \bar{u}\bar{u}^\top, \quad \Sigma_v = S - \bar{v}\bar{v}^\top, \quad K = W - \bar{v}\bar{u}^\top.$$

With

$$\alpha = \begin{cases} 1, & q = V, \\ 1 + \rho, & q = \Theta, V\Theta, \end{cases} \quad B = \begin{cases} 1 + M\rho, & q = V, V\Theta, \\ 1, & q = \Theta, \end{cases}$$

the centered long-time transport law is

$$\lim_{t \rightarrow \infty} \text{Cov}(r, u) = \frac{K}{\alpha}, \quad \lim_{t \rightarrow \infty} \text{Cov}(r, v) = \frac{M\Sigma_v + \text{Pe} K/\alpha}{B},$$

$$\text{Cov}(r(t)) = 2Dt + O(1), \quad D = \frac{\text{Cov}(r, v) + \text{Cov}(r, v)^\top}{2}, \quad D_{\text{eff}} = \frac{\text{Tr } D}{2}.$$

Thus all three non-position protocols are diffusive in centered covariance. For V , $\bar{v} = 0$, so raw MSD is also diffusive. For the two orientation-reset protocols,

$$\bar{v}_\Theta = \frac{\text{Pe}\rho}{1 + \rho}e, \quad \bar{v}_{V\Theta} = \frac{\text{Pe}\rho}{(1 + \rho)(1 + M\rho)}e.$$

Their raw MSD is therefore ballistic exactly when $\text{Pe}\rho > 0$, even though their centered fluctuations are diffusive.

Traceability: [accepted non-position solution](#).

4 S4. Exact 21-pair witness table

Assume in this section that

$$M > 0, \quad \rho > 0, \quad \text{Pe} \geq 0,$$

and that parameters and the common initial state are matched. The following three rows are disjoint and give the exact partition $12 + 6 + 3 = 21 = \binom{7}{2}$ unordered pairs.

Disjoint case	Exact pairs	Separating member of $\mathcal{R}_2$
Different position-reset indicator: 12 pairs	$P/V, P/\Theta, P/V\Theta; PV/V, PV/\Theta, PV/V\Theta; P\Theta/V, P\Theta/\Theta, P\Theta/V\Theta; PV\Theta/V, PV\Theta/\Theta, PV\Theta/V\Theta$	The position-reset member has finite stationary centered variance and bounded raw MSD; the non-position member has linearly growing centered covariance. This witness also holds at $\text{Pe} = 0$.
Same position status, different orientation-reset indicator: 6 pairs	$P/P\Theta, P/PV\Theta, PV/P\Theta, PV/PV\Theta, V/\Theta, V/V\Theta$	At $\rho > 0$, the orientation-reset member has $\bar{u} = \rho e/(1 + \rho)$ and $U = \text{diag}((\rho + 2)/(\rho + 4), 2/(\rho + 4))$; the other member has $\bar{u} = 0$ and $U = I/2$. This witness is independent of Pe .

Disjoint case	Exact pairs	Separating member of $\mathcal{R}_2$
Same position and orientation status, different velocity-reset indicator: 3 pairs	$P/PV, P\Theta/PV\Theta, \Theta/V\Theta$	For P/PV , each S_{ii} has the strictly positive difference shown below. For the other two pairs, $\bar{v}$ differs when $\text{Pe} > 0$; at $\text{Pe} = 0$, each diagonal velocity moment differs by $\rho/(2 + M\rho) > 0$.

For the P/PV pair, the all-activity witness is

$$S_{P,ii} - S_{PV,ii} = \frac{\rho [M^2\text{Pe}^2\rho + M^2\text{Pe}^2 + 2M^2\rho + 2M^2 + 3M\text{Pe}^2 + 2M\rho + 4M + 2]}{2(M+1)(M\rho+2)(M\rho+M+1)} > 0.$$

Because every unordered pair occurs in exactly one row, these witnesses prove

$$\boxed{\mathcal{R}_2(X; M, \text{Pe}, \rho) \neq \mathcal{R}_2(Y; M, \text{Pe}, \rho) \quad \text{for every } X \neq Y}$$

on the stated finite-parameter domain. The conclusion is full-record distinguishability, not practical identifiability from finite noisy data.

Traceability: [accepted exhaustive proposition](#).

5 S5. Restricted-identity ledger

The following exact equalities concern a named sector, contraction, condition, resonance, or limit. None is equality of the full records. Except for the full internal-process identities in S5.1, stationary or long-time quantities are meant where a time argument is suppressed.

5.1 S5.1 Internal-sector identities

For every finite physical parameter point,

$$PV \equiv V, \quad P\Theta \equiv \Theta, \quad PV\Theta \equiv V\Theta \quad \text{on the full internal } (v, u) \text{ process.}$$

Position reset is absent from the (v, u) generator. Hence all internal means, S, W, U , centered internal blocks, and higher internal statistics match for each pair. Their spatial records do not match: the left member is localized and the right member transports.

5.2 S5.2 Contracted-observable degeneracies

For $PV/PV\Theta$, the named scalar triple $(\text{TrS}, \text{TrC}, \text{TrR})$ agrees exactly:

$$\boxed{\text{Tr } S_{PV} = \text{Tr } S_{PV\Theta}, \quad \text{Tr } C_{PV} = \text{Tr } C_{PV\Theta}, \quad \text{Tr } R_{PV} = \text{Tr } R_{PV\Theta}.}$$

The equality follows from $\text{Tr } U = 1$. It does not extend to centered spatial variance; in the active positive-rate interior,

$$\text{Tr Cov}_{PV}(r) - \text{Tr Cov}_{PV\Theta}(r) = \frac{\text{Pe}^2}{(1 + \rho)^2(1 + M\rho)^2} > 0.$$

For $V/V\Theta$,

$$\boxed{\text{Tr } S_V = \text{Tr } S_{V\Theta}}$$

for all finite physical parameters. At $\rho > 0$, orientation moments separate the records; in the active interior, $V\Theta$ also drifts and $D_{\text{eff},V} > D_{\text{eff},V\Theta}$.

5.3 S5.3 Resonant identity

For $P/P\Theta$, each member of the named triple $(\text{Tr}S, \text{Tr}C, \text{Tr}R)$,

$$\boxed{(\text{Tr } S, \text{Tr } C, \text{Tr } R)}$$

has a protocol difference proportional to $\text{Pe}^2(M\rho - 1)$. The triple therefore agrees at the passive endpoint $\text{Pe} = 0$ and on the active resonance

$$\boxed{M\rho = 1.}$$

The full records remain distinct. For example, at $M = \rho = 1$ and $\text{Pe} = 6/5$, $\bar{u}_x$ is 0 for P and $1/2$ for $P\Theta$, while $\bar{r}_x$ is 0 and $3/5$, respectively.

5.4 S5.4 Conditional identities

At $\text{Pe} = 0$, the translational sectors satisfy

$$P = P\Theta, \quad PV = PV\Theta, \quad V = V\Theta,$$

but orientation outputs still separate each pair for $\rho > 0$.

There is also an accepted analytic-only curve on which V and Θ have the same centered diffusion tensor:

$$\boxed{\rho = \frac{1}{2}, \quad \text{Pe}^2 = \frac{27M(M+6)(3M+2)}{2(44-9M-80M^2-12M^3)}, \quad 0 < M < M_\star}$$

where $M_\star$ is the unique positive root of $44-9M-80M^2-12M^3 = 0$ ($M_\star = 0.658170892\dots$). The denominator is positive on the stated domain. This is not a full-record identity: Θ has nonzero mean drift and ballistic raw MSD, whereas V has zero drift and diffusive raw MSD. The curve is an exact analytic identity and was not part of the sampled numerical validation grid.

5.5 S5.5 Limiting identities

At fixed $\rho > 0$ and $\text{Pe} \geq 0$, first derive the exact finite- M stationary position moments (for position-reset maps) or long-time drift/diffusion coefficients (for non-position maps), and then take the position-output limit $M \rightarrow 0^+$. In this order,

$$P = PV, \quad P\Theta = PV\Theta, \quad \Theta = V\Theta \quad \text{in their position-process outputs.}$$

At $\rho = 0$, the jump term vanishes and all seven protocols reduce to the same free process. The $M \rightarrow 0^+$ statement does not assert convergence of the velocity record, a transient path-space topology, the reverse order, or a joint reset-rate limit. Equal rare- or frequent-reset leading coefficients are identities only of those named coefficients, not of the finite-rate records.

Traceability: [accepted classification and identity ledger](#) and [accepted proposition](#).

6 S6. Mandatory baselines and limits

6.1 S6.1 Complete-reset baseline

For $PV\Theta$, the stationary means reduce to

$$\bar{u} = \frac{\rho}{1+\rho}e, \quad \bar{v} = \frac{\rho \text{Pe}}{(1+\rho)(1+M\rho)}e, \quad \bar{r} = \frac{\text{Pe}}{(1+\rho)(1+M\rho)}e.$$

The complete component solution from S3.1 reproduces the accepted complete-reset inertial baseline.

6.2 S6.2 Zero-rate and rare-reset limit

For every position-reset protocol,

$$\lim_{\rho \rightarrow 0^+} \text{Tr } S = \frac{2}{M} + \frac{\text{Pe}^2}{1+M},$$

$$\lim_{\rho \rightarrow 0^+} \rho \text{Tr } R = 4 + 2\text{Pe}^2 = 4D_{\text{eff,free}}, \quad D_{\text{eff,free}} = 1 + \frac{\text{Pe}^2}{2}.$$

For each non-position protocol,

$$\lim_{\rho \rightarrow 0^+} \bar{v} = 0, \quad \lim_{\rho \rightarrow 0^+} \text{Tr } S = \frac{2}{M} + \frac{\text{Pe}^2}{1+M},$$

$$\lim_{\rho \rightarrow 0^+} D = \left(1 + \frac{\text{Pe}^2}{2}\right) I.$$

The order matters for position resetting. At each fixed $\rho > 0$, first taking $t \rightarrow \infty$ produces a stationary position law; its variance then diverges as $\rho \rightarrow 0^+$. Setting $\rho = 0$ first removes resetting and produces free diffusion rather than a stationary law. The scaled limit above connects these two descriptions without asserting that the two limits commute.

6.3 S6.3 Passive underdamped limit

At $\text{Pe} = 0$, orientation reset drops out of translational moments. For position-reset protocols,

class	S	R
$P = P\Theta$	I/M	$\frac{2I}{\rho(1+M\rho)}$
$PV = PV\Theta$	$\frac{2I}{M(2+M\rho)}$	$\frac{4I}{\rho(1+M\rho)(2+M\rho)}$

For non-position protocols,

$$D_\Theta = I, \quad D_V = D_{V\Theta} = \frac{2I}{(1+M\rho)(2+M\rho)}.$$

These are translational identities only; orientation records remain distinct when $\rho > 0$.

6.4 S6.4 Singular overdamped position-process limit

The $M \rightarrow 0^+$ comparison is a singular position-process limit. The limiting equation is

$$dr = \text{Pe } u dt + \sqrt{2} dW_t.$$

For the localized protocols, the accepted procedure is: solve the exact finite- M stationary position moments at fixed (ρ, Pe) , then take $M \rightarrow 0^+$. This gives

$$P = PV, \quad P\Theta = PV\Theta \quad \text{in position second moments,}$$

$$R_{\text{od}} = \frac{2I}{\rho} + \frac{2\text{Pe}^2 U}{\rho(1+\rho)},$$

$$\bar{r} = \begin{cases} 0, & P, PV, \\ \text{Pe } e/(1+\rho), & P\Theta, PV\Theta, \end{cases} \quad \text{Cov}(r) = R_{\text{od}} - \bar{r}\bar{r}^\top,$$

where $U = I/2$ without orientation reset and

$$U = \text{diag} \left(\frac{\rho+2}{\rho+4}, \frac{2}{\rho+4} \right)$$

with orientation reset.

For the non-position protocols, compute the finite- M diffusion tensor first and then take $M \rightarrow 0^+$:

$$D_V \longrightarrow \left(1 + \frac{\text{Pe}^2}{2}\right) I, \quad D_\Theta \longrightarrow I + \frac{\text{Pe}^2}{1+\rho} \Sigma_u, \quad D_{V\Theta} \longrightarrow I + \frac{\text{Pe}^2}{1+\rho} \Sigma_u.$$

Velocity is not an independent state at $M = 0$. These statements therefore assert convergence of the displayed position outputs, not convergence of the finite-inertia velocity record. No direct substitution $M = 0$ is made in a velocity formula. Any joint limit involving M , ρ , or $t \rightarrow \infty$ must state its order; the results above do not assert that those limits commute.

6.5 S6.5 Frequent-reset asymptotics

For position-reset protocols,

$$\begin{aligned}\lim_{\rho \rightarrow \infty} \rho^2 E_P |r|^2 &= \frac{4}{M} + \frac{2\text{Pe}^2}{1+M}, \\ \lim_{\rho \rightarrow \infty} \rho^2 E_{P\Theta} |r|^2 &= \frac{4}{M} + 2\text{Pe}^2, \\ \lim_{\rho \rightarrow \infty} \rho^3 E_{PV} |r|^2 &= \lim_{\rho \rightarrow \infty} \rho^3 E_{PV\Theta} |r|^2 = \frac{8}{M^2}.\end{aligned}$$

For $P\Theta$, $\lim_{\rho \rightarrow \infty} \rho^2 |\bar{r}|^2 = \text{Pe}^2$, so its centered variance coefficient is $4/M + \text{Pe}^2$.

For non-position protocols,

$$\begin{aligned}\lim_{\rho \rightarrow \infty} \rho^2 D_{\text{eff},V} &= \frac{\text{Pe}^2 + 4}{2M^2}, \\ \lim_{\rho \rightarrow \infty} D_{\text{eff},\Theta} &= 1, \quad \lim_{\rho \rightarrow \infty} \bar{v}_\Theta = \text{Pe} e, \\ \lim_{\rho \rightarrow \infty} \rho^2 D_{\text{eff},V\Theta} &= \frac{2}{M^2}, \quad \lim_{\rho \rightarrow \infty} \rho \bar{v}_{V\Theta} = \frac{\text{Pe}}{M} e.\end{aligned}$$

For every fixed finite $\rho > 0$, Θ and $V\Theta$ have a ballistic raw-MSD term whenever $\text{Pe} > 0$. The vanishing $V\Theta$ drift coefficient as $\rho \rightarrow \infty$ does not retroactively change that finite-rate classification. This is another noncommuting classification/asymptotic distinction: a leading coefficient may vanish in a parameter limit although the finite-parameter growth exponent remains ballistic.

7 S7. Claim boundary

The exact result is full-record second-order distinguishability of all seven nonempty reset maps for $M > 0$, $\rho > 0$, and $\text{Pe} \geq 0$, together with a scope-labeled ledger of information loss under internal-sector restriction, scalar contraction, special parameter conditions, resonance, and singular limits. Trace equality is not matrix equality; raw-MSD equality is not centered-covariance equality; and none of the restricted identities is full protocol equivalence.

The supplement makes no practical-identifiability claim and introduces no higher-order or distributional claim.

8 S8. Source crosswalk

- Model, reset maps, required observables, identity labels, and mandatory limits: [scientific contract](#).
- Exact generator, ordered redundant frame, and fail-closed closure construction: [generator.py](#).
- Position-reset stationary solution and limits: [T-020](#).
- Non-position centered transport law and limits: [T-021](#).
- Full-record proposition and exhaustive separating proof: [T-030](#).

- Exact classification, restricted identities, resonance, and tuned diffusion curve: [V-020](#).